Weighted ColorMorphology Feature Fusion for Tuberculosis Bacilli Detection from Cytopathology Images

Parshiv Kapoor*
Department of Bioscience and bioengineering
Indian Institute of Technology Roorkee
Roorkee, Uttarakhand, India
parshiv_k@bt.iitr.ac.in

Yash Bansal*
Department of Bioscience and bioengineering
Indian Institute of Technology Roorkee
Roorkee, Uttarakhand, India
yash_b@bt.iitr.ac.in

Deychen Myes*
Department of Pathology
All India Institute of Medical Sciences, New Delhi
New Delhi, New Delhi, India
myesdeychen@aiims.gov.in

Sherrin Jacob
Department of Pathology
All India Institute of Medical Sciences, New Delhi
New Delhi, New Delhi, India
sherrinjacob@hotmail.com

Venkateswaran K. Iyer
Department of Pathology
All India Institute of Medical Sciences, New Delhi
New Delhi, New Delhi, India
iyervenkat@aiims.edu

Lavleen Singh†
Department of Pathology
All India Institute of Medical Sciences, New Delhi
New Delhi, New Delhi, India
drlavleen_singh@aiims.edu

Aparajita Khan†
Department of Computer Science and Engineering
Indian Institute of Technology (BHU) Varanasi
Varanasi, Uttar Pradesh, India
aparajita.cse@iitbhu.ac.in

Partha Pratim Roy
Department of Computer Science and Engineering
Indian Institute of Technology (ISM) Dhanbad
Dhanbad, Jharkhand, India
parthapratim@iitism.ac.in

Abstract

Accurate identification of Acid-Fast Bacilli in Ziehl–Neelsen–stained cytopathology slides is critical for tuberculosis screening but is challenged by high annotation costs, staining variability, and severe class imbalance. We introduce MorphoChanNet, a lightweight classification framework trained on 100 annotated and 100 negative Field of View images from high-resolution cytopathology smear whole-slide image segments obtained from Tubercular lymphadenopathy patients at All India Institute of Medical Sciences, India. To capture fine-grained bacillary structures, our framework operates on zoomed-in image patches, enabling localized learning. The proposed MorphoChanNet model leverages a six-channel input composed of lightness, saturation, and value color channels, along with their corresponding morphological patterns extracted using Sobel edge maps, selected based on reconstruction fidelity. These multi-channel features are fused via a residual attention-based classification network with reconstruction-weighted channel fusion and a focal loss variant penalizing false positives. On our curated dataset, MorphoChanNet achieved over 97% F1-score, outperforming RGB-based baselines and deeper vision backbones, while providing interpretability through Gradient Class Activation Maps to localize bacilli via discriminative channel propagation.

Keywords

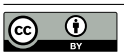
Tuberculosis Diagnosis, Acid Fast Bacilli Detection, Histopathology Image Classification, Channel Weighted Attention

ACM Reference Format:
Parshiv Kapoor, Yash Bansal, Deychen Myes, Sherrin Jacob, Venkateswaran K. Iyer, Lavleen Singh, Aparajita Khan, and Partha Pratim Roy. 2025. Weighted ColorMorphology Feature Fusion for Tuberculosis Bacilli Detection from Cytopathology Images. In *Indian Conference on Computer Vision, Graphics, and Image Processing (ICVGIP 2025), December 17–20, 2025, Mandi, India.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3774521.3774621

1 INTRODUCTION

Tuberculosis is a contagious infectious disease caused by the bacterium Mycobacterium tuberculosis, primarily affecting the lungs but capable of impacting other organs. According to the WHO Global Tuberculosis Report 2024 [28], there were an estimated 10.8 million new cases of tuberculosis worldwide in 2023. In the same year, around 1.25 million people died from tuberculosis The accurate detection of Acid-Fast Bacilli (AFB) in Ziehl–Neelsen–stained (ZN-stained) cytopathology slides is a critical step in diagnosing tuberculosis [26]. This task is traditionally performed manually

*These authors contributed equally.
†Corresponding authors. Email: aparajita.cse@iitbhu.ac.in



This work is licensed under a Creative Commons Attribution 4.0 International License.
ICVGIP 2025, Mandi, India
© 2025 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1930-1/25/12
https://doi.org/10.1145/3774521.3774621

by trained pathologists, who inspect large, high-resolution Whole-Slide Images (WSIs) under a microscope to identify the presence of AFB. However, manual review is time-consuming, labor-intensive, and prone to fatigue-related errors.These challenges make the process inefficient and difficult to scale in high-burden clinical environments. For automatic detection of AFB, several challenges must be addressed: their tiny size occupying only a small fraction of the high-resolution Field of View (FoV) images, combined with the cluttered cytology background of cells, artifacts, and debris that mimic bacilli, often leads to false positives [29]. Severe class imbalance, due to the rarity of AFB-positive patches, further complicates model training, while staining variability across slides introduces domain shifts that hinder generalization.

To address this, several deep learning pipelines have been proposed that operate on annotated FoVs extracted from WSIs. These approaches typically rely on patch-based classification or object detection frameworks trained on large-scale datasets [17], [12], [23], [7]. Chang et al. [2] employed morphological operations, followed by feature-based characterization, and used SVM-based classification. With the advent of convolutional neural networks (CNNs), several deep learning frameworks have been applied to this task. For instance, López et al. [17] trained a CNN on small bright-field patches for TB detection, experimenting with RGB, LAB, and grayscale input variants, while Kant and Srivastava [14] similarly trained a five-layer CNN on patch-based inputs extracted at zoom levels to classify bacterial presence. It is noted that the foundation for most of these techniques is based solely on simple CNNs that cover only a tenuous connection to the domain knowledge.

Recent advancements have leveraged deeper and wider convolutional architectures such as ResNet [11], InceptionV3 [25], and DenseNet [13]-often with transfer learning-for bacterial image classification tasks [23] , [12]. These approaches are capable of isolating and identifying small bacterial regions with high accuracy, effectively learning discriminative representations directly from raw microscopy images. This has led to significant improvements in patch-level classification performance. However, these models introduce considerable computational complexity and require large-scale annotated data for effective fine-tuning. This remains a key bottleneck, as generating high-quality annotations demands deep domain expertise and is both time-consuming and labor-intensive. Also, existing methods largely treat multi-channel microscopy images as undifferentiated RGB inputs and overlook the potential of color channel-specific cues and adaptive channel weighting to improve model interpretability and performance. In particular, explicitly modeling morphology-relevant color decompositions and weighting channels based on their structural saliency remains underexplored.

Therefore, we propose a novel morphology-aware weighted channel-based lightweight classification framework **MorphoChanNet** is introduced to address the staining variability and capture fine-grained bacterial morphological structures.MorphoChanNet enables the model to exploit staining-independent features through channel selection while also incorporating zoom-level patch inputs to capture fine morphological details using limited data. The input channels used are lightness, saturation, and hue channels alongside their corresponding Sobel edge maps, making a six-channel representation. This excels at differentiating true bacterial forms from common, similar-looking stained artifacts, leveraging a false-positive penalizing loss function, delivering high accuracy with only a small, expertly annotated dataset. The main contributions of this work are:

- A novel channel-based weighting mechanism and optimized color-space selection for enhancing AFB detection.
- An enhanced version of Convolutional Block Attention Module (CBAM) [27] using residual connections to preserve fine-grained bacterial structures.
- A curated, expert-annotated dataset of light-field smear microscopy patches for AFB classification and a zoomed-in patch training strategy

We demonstrate the effectiveness of our model through comprehensive ablation studies and comparisons with state-of-the-art baseline approaches on our curated dataset.

2 RELATED WORKS

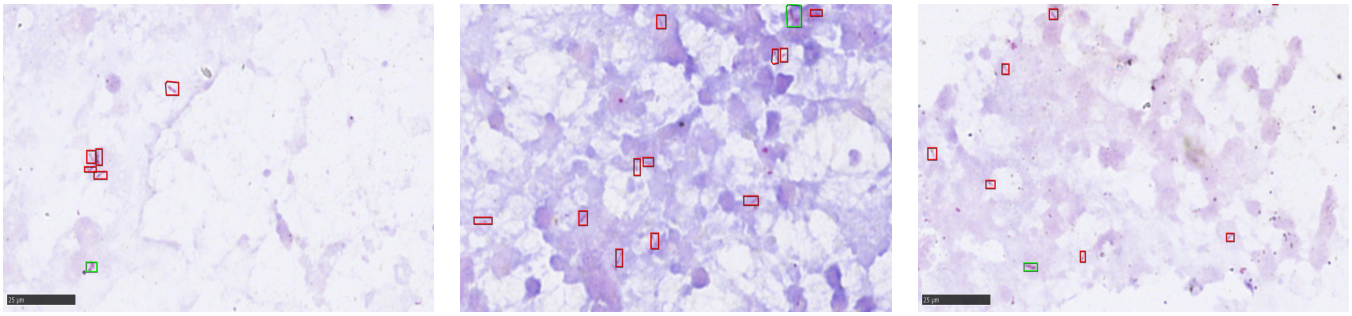
This section presents a review of existing research on bacilli detection and image classification deep learning techniques based on principles of color space optimization.

2.1 Deep Learning based Acid Fast Bacilli detection

Recent TB microscopy studies have employed deep CNN backbones, often pre-trained on large-scale image datasets, to classify smear image patches or FoVs. For example, Kant and Srivastava et al. [14] trained a five-layer CNN on zoomed bacterial patches. Karlita et al [15] employed an attention-based multi-scale CNN to automatically detect Mycobacterium tuberculosis in ZN-stained sputum smear images by combining spatial attention modules with feature maps extracted at multiple scales. However, the approach doesn't leverage channel-wise morphological or color features for classification. Other works have adopted well-established deep architectures: Samuel and Kanna extracted features using a pre-trained InceptionV3 and classified FoVs using an SVM [23], Ahmed et al. [1] used InceptionV3 as a feature encoder for multiclass bacterial identification, Rachmad et al. [18] utilized ResNet101 to extract deep features, then trained an SVM on the embeddings to discriminate TB bacteria.Hu et al. [12]partitioned WSIs into subgraphs and applied ResNet [11], InceptionV3 [25], and DenseNet [13] (all with transfer learning) to classify them. In each case, for classification, these approaches rely on significantly larger, bounding box-annotated datasets (ranging from hundreds to thousands of images), and typically employ deeper, wider, and more complex architectures such as Inception, DenseNet, and ResNet variants. Several recent works employ segmentation-based methods for detection [3, 9, 21]; however, our approach focuses solely on accurate classification without requiring pixel-wise annotations.

2.2 Color-Space based Classification

Color space transformations enhance the visibility of subtle morphological cues, particularly in small object detection and medical imaging. Studies have shown that non-RGB spaces like LAB, HSV, and YCbCr improve shape and edge sensitivity [8, 19]. In



(a) Image 1 (b) Image 2 (c) Image 3

Figure 1: Expert Annotated ZN stained images used for dataset construction.

bacterial classification, leveraging such transformations helps disentangle stain variation from structural features, enabling better morphological representation [24].Such methods reduce bias toward specific staining protocols and offer a robust foundation for downstream classification.Díaz-Huerta et al.[5] converted RGB images to HSV color space for simple preprocessing and applied a basic Artificial Neural Network for pixel-wise bacilli segmentation, yielding modest detection gains but failing to capture fine morphological cues due to the limited representational capacity of simple CNNs.Chithra and Jagatheeswari [4] transformed images to LUV color space, used adaptive thresholding, handcrafted feature extraction (coverage, density, histograms, geometry, texture), PCA, and a fractional crow search–guided CNN for detection and severity scoring. While comprehensive, this multi-stage pipeline incurs a high computational.López et al. [17] trained a CNN on small bright-field patches for TB detection, experimenting with RGB, LAB, and grayscale input variants.Lopez et al.[17] employ patch-based training across multiple color-space inputs (RGB, R–G, and grayscale), but their uniform treatment of all channels-without incorporating edge-aware cues and channel-wise attention, prevents the model from prioritizing morphology-relevant features. This limitation, inherent to conventional CNNs lacking adaptive channel mechanisms, forfeits the opportunity to learn robust, morphology-aware representations of Mycobacterium tuberculosis.

Our Contribution - We propose an AFB detection pipeline that uses channel-specific features to classify bacteria and understand the morphological features. Our approach begins by decomposing microscopy images into a six-channel representation, selectively including lightness (L), saturation (S), and value (V) components alongside Sobel-based edge maps. These channels were chosen based on their minimal reconstruction error and their ability to preserve bacterial morphology under stain variability. In our pipeline, we train a lightweight classifier model with channel-wise weightage to inputs, where more informative channels are emphasized, and a false-positive-penalized focal loss to achieve accurate classification of images based on AFB's presence. To further enhance interpretability, we integrate Gradient Class Activation Map (Grad-CAM) visualizations to highlight salient regions contributing to predictions, enabling better trust in automated decisions.

3 DATASET

We begin with a curated collection of 200 high-resolution WSIs which consisted of 100 positives and 100 negatives, each with a spatial resolution of 1368×776 pixels. These slides are ZN-stained cytopathology smears collected from clinical samples as illustrated in Figure 1. The staining intensity varies significantly across slides due to differences in sample preparation, fixation, and scanning conditions. All images were manually annotated by domain experts, with red bounding boxes marking regions containing visible AFB. Each image in the dataset contains at least one confirmed instance of the presence of bacteria.

3.1 Dataset Preprocessing

To construct the positive sample set, we employed a sliding window approach to extract 80×80 patches from the images. A patch was labeled as positive if it contained at least 70% of the area of an expert-annotated bounding box corresponding to a bacterial region. This procedure resulted in a total of 500 positive samples.

To construct a robust and diverse negative dataset, we employed a structured sampling strategy across the 100 annotated and color-normalized histopathology slides (1368×776 pixels each). A sliding window was used to extract non-overlapping 80×80 patches in which we selected visually challenging negatives to enhance model discrimination. We also leveraged the use of 100 images which were fully negative (no presence of bacteria) :

- Unstained Regions: Patches with >95% white pixels to represent background or empty tissue.
- Staining Transition Zones: Boundary areas with 60–95% white space and strong gradient activity.
- Overstained Areas and Artifacts: Dense dye regions that may resemble bacterial clusters.
- Bacteria-Mimicking Artifacts: Noise patterns or smudges with structural similarities to bacteria.

This targeted strategy resulted in approximately 10,000 negative patches. To address class imbalance, the positive set was upsampled from 500 to 8,000 samples through label-preserving augmentations, including random 90° rotations, horizontal and vertical flips, mild Gaussian blurring, and additive noise. The resulting data distributions after augmentation are summarized in Table 1. These augmentations preserved bacterial morphology while improving model generalization to appearance variations.

Table 1: Distribution of samples before and after augmentation.

Phase	Positive	Negative	Ratio
Before Aug.	500	10,000	1:20 (5%)
After Aug.	8,000	10,000	1:1.25 (80%)

All patches were then enhanced using Contrast Limited Adaptive Histogram Equalization [10] with a clip limit of 2.0. This adaptive contrast enhancement technique improves local detail visibility, particularly in regions with uneven staining, without amplifying noise artifacts.

Finally, each patch image was converted into a six-channeled representation by concatenating three selected color channels (based on reconstruction loss; see Section 4.1) with their corresponding Sobel edge maps.Figure 2 illustrates the individual color channels and their associated edge maps for representative patch images.

4 METHODOLOGY

Our method combines high-resolution patch-based classification with channel-wise weighting, where the selected color channels are adaptively reweighted based on reconstruction loss computed via a pretrained U-Net [20]. To enhance spatial focus, we further incorporate a channel-wise attention mechanism. For interpretability, we employ Grad-CAM [22] through a sliding window over the full-resolution image, generating heatmaps that highlight diagnostically relevant regions.

4.1 Color Space Selection via Reconstruction Fidelity

Effective color representation is essential for detecting fine bacterial structures under variable staining conditions. To identify structurally informative visual cues, we first convert each 80×80 patch into 18 single-channel grayscale images corresponding to six color spaces: RGB (Red, Green, Blue), HSV (Hue, Saturation, Value), HLS (Hue, Lightness, Saturation), YCrCb (Luminance, Chroma-Red, Chroma-Blue), YUV (Luminance, Blue-difference, Red-difference), and LAB (Lightness, A (green–red), B (blue–yellow)).

Let the individual grayscale channels be denoted by $X^{(i)} \in \mathbb{R}^{B\times 1\times H\times W}$ for $i = 1, \ldots, 18$, where B is the batch size and $H = W = 80$. Each channel $X^{(i)}$ is passed through a dedicated frozen U-Net [20] reconstruction network $f_{\text{recon},i}$, which employs a ResNet-18 encoder pretrained on ImageNet:

$$\hat{X}^{(i)} = f_{\text{recon},i}(X^{(i)}), \quad \hat{X}^{(i)} \in \mathbb{R}^{B\times 1\times H\times W} \tag{1}$$

Reconstruction quality is measured using the Mean Squared Error (MSE) loss over spatial dimensions:

$$\ell_{\text{recon}}^{(i)} = \frac{1}{H \cdot W} \sum_{h=1}^{H} \sum_{w=1}^{W} \left(X_{[:,:,h,w]}^{(i)} - \hat{X}_{[:,:,h,w]}^{(i)} \right)^2 \tag{2}$$

Channels with the lowest reconstruction losses—indicating superior structural preservation—are selected based on Eq.(2). These include the lightness channel (L) from HLS, saturation (S), and value (V) channels from HSV.

To further emphasize bacterial morphology, we compute gradient-based edge maps using Sobel filters. For each selected channel $X^{(i)}$, horizontal and vertical gradients $G_x^{(i)}$ and $G_y^{(i)}$ are computed, and the corresponding edge map is obtained via magnitude aggregation:

$$G^{(i)} = \sqrt{\left(G_x^{(i)}\right)^2 + \left(G_y^{(i)}\right)^2} \tag{3}$$

Finally, the original channel $X^{(i)}$ is concatenated with its edge map $G^{(i)}$, forming a two-channel representation. Concatenating across all three selected color channels yields the final input tensor:

$$X_{\text{input}} \in \mathbb{R}^{B\times 6\times H\times W} \tag{4}$$

which is used for classification.The final input is a six-channel representation as shown in Eq.(4).

4.2 Model Architecture

Our proposed architecture, shown in Figure 3 is structured into three functional modules: (i) a Classifier Backbone with Residual Attention for learning discriminative representations, (ii) a Multi-Decoder Reconstruction Module that enforces channel-wise structural fidelity, and (iii) a Reconstruction-Guided Channel Weighting Mechanism that adaptively modulates input contributions based on reconstruction quality. These components are described in detail below.

4.2.1 Classifier Backbone with Residual Attention. To enhance spatial and morphological discrimination in microscopy patches, we employ channel-wise attention-aware residual blocks in the classification backbone. Specifically, we integrate an enhanced version of the CBAM [27] at each stage of the network, embedded within residual connections shown in Figure 4. This module sequentially applies channel and spatial attention, enabling the model to focus on the most informative features across both dimensions.

Given an input feature map $\mathbf{F}_{\text{in}} \in \mathbb{R}^{C\times H\times W}$, the Enhanced CBAM block enhances it via sequential channel and spatial attention mechanisms, embedded within a residual structure as follows:

1. Channel and Spatial Attention. To enhance discriminative feature learning, we employ the CBAM block [27], which sequentially applies channel and spatial attention. First, global average and max pooling are applied across spatial dimensions, followed by a shared MLP with reduction ratio r to produce the channel attention map:

$$\mathbf{M}_{\text{c}} = \sigma\left(\text{MLP}(\text{AvgPool}(\mathbf{F}_{\text{in}})) + \text{MLP}(\text{MaxPool}(\mathbf{F}_{\text{in}}))\right). \tag{5}$$

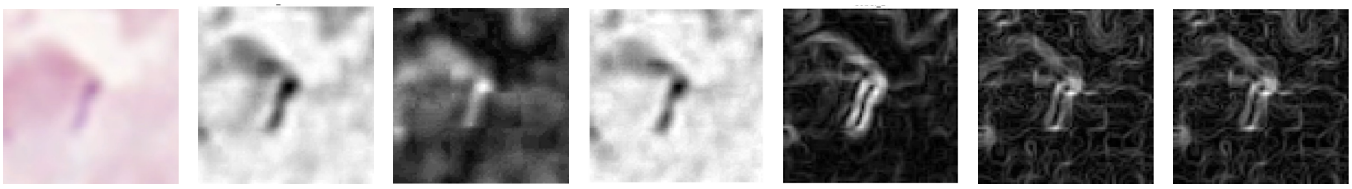
The input is refined as $\mathbf{F}_{\text{c}} = \mathbf{M}_{\text{c}} \odot \mathbf{F}_{\text{in}}$, where $\odot$ denotes element-wise multiplication broadcast across spatial dimensions. Next, average and max pooling are performed along the channel axis, concatenated, and convolved with a 7×7 kernel to generate the spatial attention map:

$$\mathbf{M}_{\text{s}} = \sigma\left(f^{7\times 7}([\text{Mean}_c(\mathbf{F}_{\text{c}}); \text{Max}_c(\mathbf{F}_{\text{c}})])\right). \tag{6}$$

The final attention-refined feature is:

$$\mathbf{F}_{\text{sa}} = \mathbf{F}_{\text{out}} = \mathbf{M}_{\text{s}} \odot \mathbf{F}_{\text{c}}. \tag{7}$$

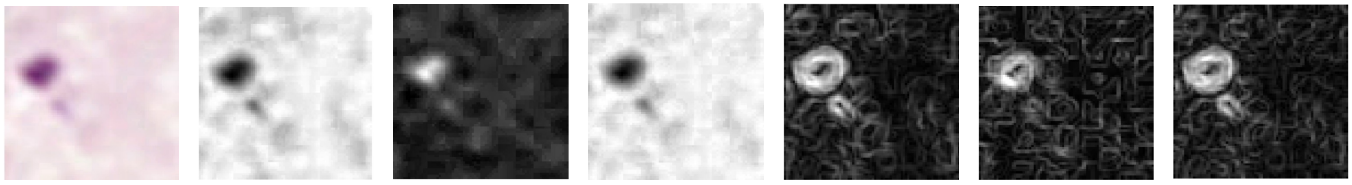
(a) Positive Sample



(a) RGB (b) L (c) S (d) V (e) Sobel L (f) Sobel S (g) Sobel V

Positive sample showing RGB, L, S, V channels and corresponding Sobel edge maps.

(b) Negative Sample



(h) RGB (i) L (j) S (k) V (l) Sobel L (m) Sobel S (n) Sobel V

Negative sample showing RGB, L, S, V channels and corresponding Sobel edge maps.

Figure 2: Comparison of positive and negative six-channeled samples. Each row illustrates the RGB image, L/S/V channels, and their Sobel edge maps.

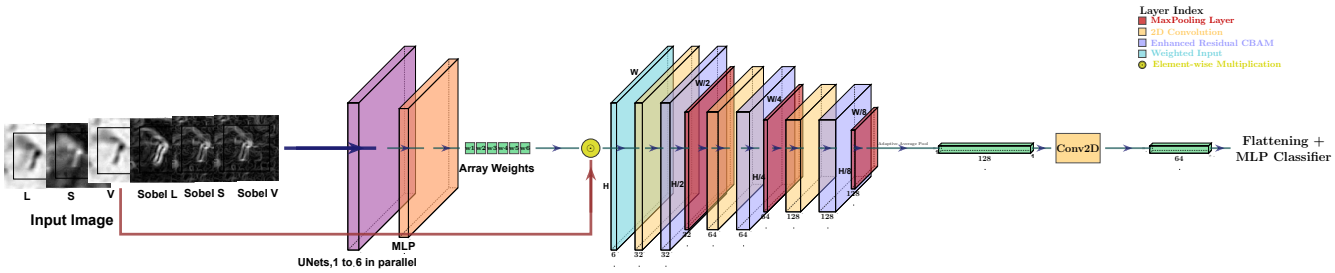


Figure 3: Overview of our proposed architecture-MorphoChanNet

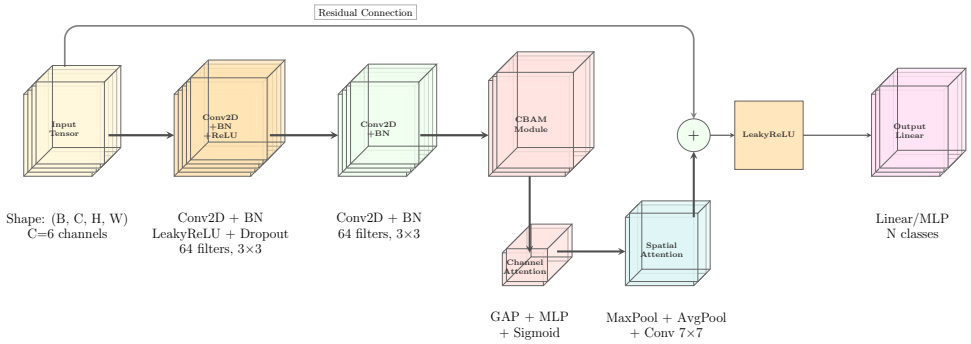


Figure 4: Enhanced Residual CBAM architecture.

2. Residual CBAM Block. We first apply a convolutional block $\mathcal{B}(\cdot)$ (Conv–BN–ReLU–Dropout–Conv–BN), then apply CBAM attention:

$$\mathbf{F}_{\text{cbam}} = \text{CBAM}\big(\mathcal{B}(\mathbf{F}_{\text{in}})\big). \tag{8}$$

Finally, the residual output is:

$$\mathbf{F}_{\text{out}} = \text{ReLU}\big(\mathbf{F}_{\text{in}} + \mathbf{F}_{\text{cbam}}\big). \tag{9}$$

In this formulation, channel attention (Eq.(5)), adaptively emphasizes inter-channel dependencies, while spatial attention, (Eq.(6)) highlights salient spatial regions within each feature map, culminating in the refined output (Eq.(7)). Our proposed residual CBAM block, unlike standard CBAM which lacks identity mapping, preserves low-level texture and fine-grained structure critical for discriminating bacilli morphologies. The residual connection also facilitates gradient flow, improving training stability and convergence. Compared to transformer-style self-attention over tokens ($\mathbb{R}^{B\times N\times C}$), CBAM uses lightweight convolutional primitives that retain spatial locality, making it computationally efficient and well-suited for biomedical imaging scenarios where preserving spatial detail is crucial.

4.2.2 Multi-Decoder Reconstruction Module. In parallel, the model includes a set of six frozen U-Net [20] reconstruction networks—one for each input channel. Each U-Net follows an encoder-decoder architecture with a ResNet-18 [11] backbone pretrained on ImageNet. These networks reconstruct their respective channels independently and are kept frozen during classifier training.

Given an input image patch $X \in \mathbb{R}^{B\times 6\times 80\times 80}$, where B denotes the batch size and the six channels correspond to distinct color and where $H = W = 80$ denote the spatial height and width of each channel, the channel-wise reconstruction losses for a batch are collected into a vector:

$$\boldsymbol{\ell}_{\text{recon}} = [\ell^{(1)}_{\text{recon}}, \ell^{(2)}_{\text{recon}}, \ldots, \ell^{(6)}_{\text{recon}}] \in \mathbb{R}^{B\times 6}. \tag{10}$$

4.2.3 Reconstruction-Guided Channel Weighting. To prioritize channels capturing bacilli more distinctly, we use reconstruction difficulty as a proxy for semantic saliency. The per-channel reconstruction losses $\boldsymbol{\ell}_{\text{recon}} \in \mathbb{R}^{B\times 6}$ are passed through a shared MLP with sigmoid activation:

$$\mathbf{s} = \sigma\left(\text{MLP}_\theta(\boldsymbol{\ell}_{\text{recon}})\right) \in \mathbb{R}^{B\times 6}, \tag{11}$$

where σ is the element-wise sigmoid. Channels with lower reconstruction error receive higher weights by inverting and normalizing the outputs:

$$w_i = \frac{1 - s_i}{\sum_{j=1}^{6}(1 - s_j)} \quad \text{for } i = 1, \ldots, 6, \tag{12}$$

yielding $\mathbf{w} \in \mathbb{R}^{B\times 6}$, reshaped as $\mathbf{w}_{\text{reshaped}} \in \mathbb{R}^{B\times 6\times 1\times 1}$. The input $X \in \mathbb{R}^{B\times 6\times H\times W}$ is modulated by

$$X_{\text{weighted}} = X \odot \mathbf{w}_{\text{reshaped}}, \tag{13}$$

and passed to the classifier:

$$y = f_{\text{cls}}(X_{\text{weighted}}), \tag{14}$$

enabling adaptive emphasis on morphologically salient channels through reconstruction-aware soft attention.

4.3 Loss Function

To mitigate the pronounced class imbalance between background and bacteria-containing patches, and to reduce the clinical risk posed by false negatives, we adopt a modified variant of the Focal Loss [16] as our training objective. We emphasize sensitivity to the positive class by setting the class-balancing factor $\alpha = 0.7$, which up-weights bacteria-containing patches in the loss computation.

Beyond the standard Focal Loss, we introduce an asymmetric penalty for false positives by applying a multiplicative penalty factor $\lambda_{\text{FP}} = 1.75$ to incorrect positive predictions. This encourages the model to be more selective in its predictions, effectively suppressing artifact-driven responses while still maintaining sensitivity to true bacterial instances at the patch level. The modified loss is defined as:

$$\mathcal{L}_{\text{mod}}(p, y) = -\alpha\, y(1-p)^\gamma \log p - \lambda_{\text{FP}}(1-\alpha)(1-y)p^\gamma \log(1-p) \tag{15}$$

where $p \in (0, 1)$ denotes the predicted probability, $y \in \{0, 1\}$ is the ground-truth label, $\gamma = 0.5$ is the focusing parameter, and $\lambda_{\text{FP}} = 1.75$ explicitly penalizes false positives more heavily than false negatives.The loss formulation as shown in Eq.(15) encourages the model to focus more on challenging or uncertain bacterial regions-such as faint bacilli that are visually similar to the background-while still controlling the tendency for false positives. In practice, this helped improve specificity without reducing recall (see Table 4).

Although our model is trained only with image-level binary labels, we extract spatial localization cues using Grad-CAM. This method backpropagates gradients from the classification score to the final convolutional layer to generate a coarse importance map. For each input image, Grad-CAM produces a heatmap $L^{\text{Grad-CAM}} \in \mathbb{R}^{H\times W}$ over each sliding window of 80×80 of the full-resolution image. These localized activation maps not only enhance model transparency for clinical workflows, but also confirm that the network learns to attend to both the spatial morphology and localization of the bacteria.

5 RESULTS

We evaluate our proposed reconstruction-weighted attention model on a cytopathology dataset with variable staining obtained from Tubercular lymphadenopathy patients at All India Institute of Medical Sciences, India to detect sparsely distributed AFB. This section outlines our results , ablation studies,training protocol, experimental settings, and key hyperparameters.

5.1 Experimental Setup

The network was trained on the proposed *six-channeled* input representation, constructed via edge-enhanced fusion of multiple color spaces, as detailed in Section 4.2. The dataset was split into training, validation, and test sets using a 3:1:1 ratio and for inference a fixed classification threshold of 0.4 was used.All models were trained using the AdamW optimizer. We set the learning rate to 3×10^{-4}, weight decay to 1×10^{-3}, and trained for 20 epochs. To ensure numerical stability and prevent exploding gradients, gradient clipping was applied with a maximum norm of 1.0.

Experiments were conducted on an NVIDIA GeForce RTX 3090 GPU, the average dataset preprocessing time per whole-slide image of size 1368×776 pixels was approximately 0.564 s, and the average

Table 2: Comparative performance analysis of different color spaces and morphological operators for edge detection

Color Space	None & Sobel						Morph & Canny					
	Edge	Precision	Recall	Specificity	F1-score	Accuracy	Edge	Precision	Recall	Specificity	F1-score	Accuracy
RGB	None	90.48	97.08	92.77	93.66	94.56	Morph	93.71	97.81	95.35	95.71	96.37
	Sobel	93.85	98.80	95.38	95.10	96.10	Canny	96.79	95.38	97.76	96.08	96.77
HLS	None	94.81	97.81	96.21	96.29	96.88	Morph	94.47	**99.70**	95.87	97.04	97.48
	Sobel	94.88	99.27	96.21	97.03	97.48	Canny	94.52	96.60	96.04	95.55	96.27
HSV	None	92.52	99.27	94.32	95.77	96.37	Morph	92.45	98.30	94.32	95.28	95.97
	Sobel	92.71	99.03	94.50	95.75	96.37	Canny	93.41	96.59	95.18	95.00	95.77
YCrCb	None	90.67	99.27	92.77	94.77	95.46	Morph	92.29	99.05	94.15	95.54	96.17
	Sobel	94.68	98.30	95.88	96.84	96.48	Canny	92.74	96.35	94.66	94.50	95.36
YUV	None	90.50	99.51	92.60	94.79	95.46	Morph	91.89	99.27	93.80	95.44	96.07
	Sobel	94.00	99.03	95.52	96.45	96.98	Canny	89.43	98.78	91.74	93.87	94.66
LAB	None	91.48	99.27	93.46	95.22	95.87	Morph	89.25	99.03	91.57	93.89	94.65
	Sobel	94.24	99.51	95.70	96.80	97.28	Canny	92.54	96.59	94.49	94.52	95.36
LSV (Ours)	None	92.50	99.03	94.32	95.65	96.27	Morph	92.43	98.05	94.32	95.16	95.95
	Sobel	**96.58**	**99.40**	**96.87**	**97.77**	**97.65**	Canny	94.63	98.54	96.04	96.54	97.08

Table 3: Ablation of architectural and loss function components

Method	Precision	Recall	Specificity	F1 Score	Accuracy	PR AUC
LSV + BCE	94.72	96.89	96.10	95.78	96.42	98.85
LSV + Focal Loss	94.91	97.64	96.28	96.31	96.87	99.02
LSV + Focal Loss + Enhanced CBAM	95.22	98.54	96.35	96.91	97.29	99.21
MorphoChanNet (random weights)	93.84	98.72	95.18	96.21	96.58	98.94
MorphoChanNet (with reconstruction guided weighting)	**96.58**	**99.40**	**96.87**	**97.77**	**97.65**	**99.32**

inference time per image was 0.542 s. The proposed model has a total of 0.49 M trainable parameters.

5.2 Ablation Studies

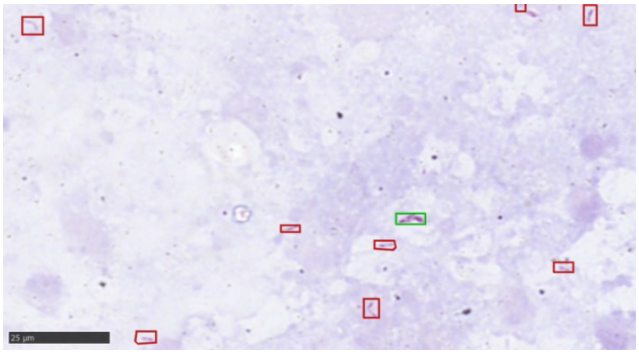
5.2.1 Color Space and Edge Enhancement Ablation. We conducted an extensive ablation study spanning both color space transformations and classical (non-deep learning) edge enhancement techniques. Specifically, we evaluated six widely used color representations: RGB, HLS, HSV, YCrCb, YUV, and LAB. In parallel, we explored four edge enhancement heuristics: (1) no edge detection (standard 3-channel image), (2) Sobel filters, which compute intensity gradients along spatial axes, (3) Morphological gradient, which highlights contour transitions via dilation–erosion differences, and (4) Canny edge detection.These experiments were run under identical training settings to isolate the effect of input representation on model performance. The resulting classification metrics are reported in Table 2. Among the tested configurations, the use of LSV (Lightness,Saturation,Value) channels together with Sobel edge maps showed improved Accuracy and F1 Score. This setup also provided better Grad-CAM [22] overlays showing improved localization.

5.2.2 Model Architecture Ablations. To quantify the contribution of each architectural enhancement in MorphoChanNet, we performed a series of controlled ablation experiments summarized in Table 3. We begin with a baseline model using the LSV color space and Binary Cross-Entropy loss. Replacing it with Focal loss yields consistent improvements in all metrics.Adding the enhanced CBAM attention module further boosts recall and F1 score. Finally, integrating our channel-aware reconstruction-weighted fusion yields the best results, with the highest precision (96.58), recall (99.40), and F1 score (97.77).

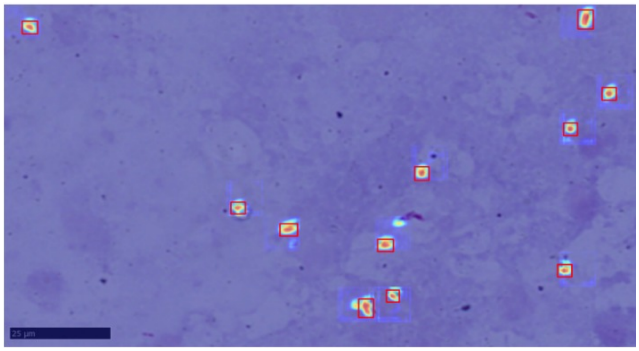
5.3 Patch-wise Classification and Inference

The model was trained for binary classification on histopathology patches to determine the presence or absence of bacteria. Table 3 shows that, on the held-out test set, MorphoChanNet achieved a recall of 99.40, along with an overall accuracy of 97.65.

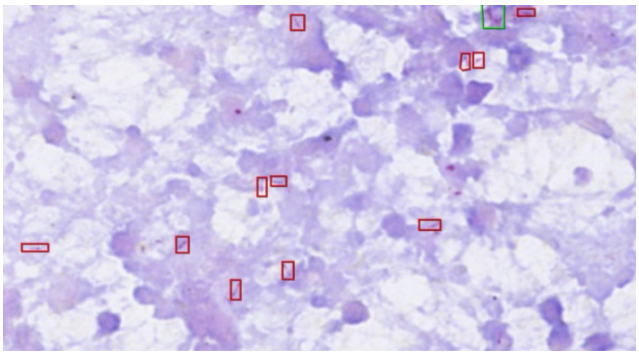
For whole-slide inference, we used a sliding window approach to extract overlapping patches of size 80 × 80 with a stride of 40 from the full-resolution slides. Each patch was then classified independently and applied Grad-CAM [22]. Qualitative examples with predicted bounding boxes using thresholding overlaid on Grad-CAM activations are shown in Figure 5. For patches predicted as positive, the gradient heatmaps were thresholded to highlight salient regions, and connected component analysis was used to generate bounding boxes around areas with strong activations. As shown in Figure 5b, the network highlights regions with morphological patterns that are associated with bacterial presence. This interpretability can help in integrating the approach into cytopathological workflows and may support future segmentation methods without requiring detailed pixel-wise annotations.



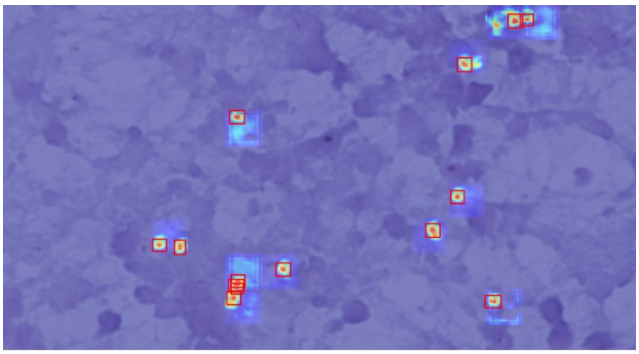
(a) Ground truth annotation showing expert-labeled AFB-positive regions.



(b) Grad-CAM activation overlays(Display Grad-CAM overlay only if activation ≥ threshold; otherwise show a fully blue image) .



(c) Ground truth annotation showing expert-labeled AFB-positive regions.



(d) Grad-CAM activation overlays (Display Grad-CAM overlay only if activation ≥ threshold; otherwise show a fully blue image)

Figure 5: Qualitative Localization on Our Dataset. Comparison of expert annotations (a, c) and Grad-CAM-based predictions (b, d) on AFB-positive FoVs.

Table 4: Comparison with Existing Approaches: 3-Channel vs 6-Channel Input

Method	3-Channel Input (Standard RGB)					6-Channel Input (RGB + HSV)				
	Prec.	Rec.	Spec.	F1	Acc.	Prec.	Rec.	Spec.	F1	Acc.
ResNet-50	80.06	93.87	81.30	86.42	86.89	91.63	95.86	93.70	93.80	94.66
ViT-B/16	**91.58**	**97.88**	**92.80**	**94.62**	**95.06**	94.63	98.54	96.04	96.54	97.08
ResNet-101	87.59	98.19	88.39	92.50	92.69	94.90	**99.51**	96.21	97.15	97.58
DenseNet-201	86.13	95.75	87.36	90.58	91.08	93.44	97.08	95.18	95.23	95.97
InceptionV3	80.77	72.39	85.12	75.61	80.09	88.08	73.72	92.94	80.26	84.98
EfficientNet-B7	76.55	82.12	80.82	79.01	81.59	83.86	83.45	88.64	83.66	86.49
DeiT	88.15	88.41	89.20	88.28	89.05	92.29	89.75	94.00	91.00	92.11
EfficientNetV2	84.73	89.28	87.45	86.94	87.61	94.01	92.13	95.30	93.06	93.89
ConvNeXt	70.12	70.85	75.42	70.48	73.18	78.86	77.88	83.30	78.36	80.89
Ours	–					**96.58**	99.40	**96.87**	**97.77**	**97.65**

5.4 Comparison with State-of-the-Art Methods

Existing state-of-the-art methods for bacterial detection predominantly leverage deep and wide architectures with transfer learning. For instance, Samuel and Kanna employed InceptionV3 as a feature extractor followed by SVM classification [14], while Ahmed et al.[1] utilized InceptionV3 for multiclass bacterial identification. Rachmad et al.[18] extracted features using ResNet101 and trained an SVM for TB discrimination, and Hu et al.[12] applied deep CNN backbones including ResNet, DenseNet, and InceptionV3 across subgraphs of WSIs. To verify the efficacy of these approaches on our dataset, we conducted ablation studies using several such backbones (Table 4) under identical training conditions with both 6 channeled and standard 3 channeled RGB input . ResNet-101 [11] achieved the highest recall (99.51), while ViT-B/16 [6] attained strong overall

performance (F1 = 96.54, Accuracy = 97.08). In particular, our proposed model achieved comparable or superior performance in most metrics (F1 = 97.17, Accuracy = 97.58) despite having a significantly shallower and more lightweight trainable architecture. This result underscores the efficiency of our design, offering strong diagnostic performance.

6 CONCLUSION

We address the challenging task of classifying ZN–stained slides for AFB detection, crucial for tuberculosis diagnosis. Our architecture employs a reconstruction-guided channel weighting mechanism that selectively emphasizes stain-invariant, morphology-relevant features, improving generalization across variable staining while preserving fine structural detail. In future work, this framework can be extended towards accurate localization of bacteria, with the potential to distinguish true bacilli from false positive artifacts by incorporating a deeper understanding of bacterial morphology.

References

[1] Tasnim Ahmed, Md Ferdous Wahid, and Md Jahid Hasan. 2019. Combining deep convolutional neural network with support vector machine to classify microscopic bacteria images. In *2019 International Conference on Electrical, Computer and Communication Engineering (ECCE)*. IEEE, 1–5.

[2] Jeannette Chang, Pablo Arbeláez, Neil Switz, Clay Reber, Asa Tapley, J Lucian Davis, Adithya Cattamanchi, Daniel Fletcher, and Jitendra Malik. 2012. Automated tuberculosis diagnosis using fluorescence images from a mobile microscope. In *International conference on medical image computing and computer-assisted intervention*. Springer, 345–352.

[3] Shuang Yee Chin, Jian Dong, Khairunnisa Hasikin, Romano Ngui, Khin Wee Lai, Pauline Shan Qing Yeoh, and Xiang Wu. 2024. Bacterial image analysis using multi-task deep learning approaches for clinical microscopy. *PeerJ Computer Science* 10 (2024), e2180.

[4] RS Chithra and P Jagatheeswari. 2020. Severity detection and infection level identification of tuberculosis using deep learning. *International Journal of Imaging Systems and Technology* 30, 4 (2020), 994–1011.

[5] Jorge Luis Díaz-Huerta, Adriana del Carmen Téllez-Anguiano, José Antonio Gutiérrez-Gnecchi, Owen Yair Colin-González, Fanny Lucía Zavala-Santoyo, and Sergio Arellano-Calderón. 2019. Image preprocessing to improve Acid-Fast Bacilli (AFB) detection in smear microscopy to diagnose pulmonary tuberculosis. In *2019 International Conference on Electronics, Communications and Computers (CONIELECOMP)*. IEEE, 66–73.

[6] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, et al. 2020. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929* (2020).

[7] Moumen El-Melegy, Doaa Mohamed, and Tarek ElMelegy. 2019. Automatic detection of tuberculosis bacilli from microscopic sputum smear images using faster R-CNN, transfer learning and augmentation. In *Iberian Conference on Pattern Recognition and Image Analysis*. Springer, 270–278.

[8] Shreyank N Gowda and Chun Yuan. 2018. ColorNet: Investigating the importance of color spaces for image classification. In *Asian conference on computer vision*. Springer, 581–596.

[9] K Greeshma and S Vishnukumar. 2025. Enhanced Tuberculosis Bacilli Detection using Attention-Residual U-Net and Ensemble Classification. *arXiv preprint arXiv:2501.03539* (2025).

[10] Yasser Radouane Haddadi, Boualem Mansouri, and Fatima Zohra Idriss Khodja. 2024. A novel medical image enhancement algorithm based on CLAHE and pelican optimization. *Multimedia Tools and Applications* 83, 42 (2024), 90069–90088.

[11] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 770–778.

[12] Mengying Hu, Yiqing Liu, Yexing Zhang, Tian Guan, and Yonghong He. 2019. Automatic detection of tuberculosis bacilli in sputum smear scans based on subgraph classification. In *2019 International Conference on Medical Imaging Physics and Engineering (ICMIPE)*. IEEE, 1–7.

[13] Gao Huang, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q Weinberger. 2017. Densely connected convolutional networks. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 4700–4708.

[14] Sonaal Kant and Muktabh Mayank Srivastava. 2018. Towards automated tuberculosis detection using deep learning. In *2018 IEEE Symposium Series on Computational Intelligence (SSCI)*. IEEE, 1250–1253.

[15] Tita Karlita, Fidisa Anindya Pastika, Nana Ramadijanti, Ratna Kusumawati, Heny Yuniarti, and Riyanto Sigit. 2024. Mycobacterium Tuberculosis Detection on Sputum Smear Microscopic Images Using Attention-Based Multi-Scale Convolutional Neural Network. In *2024 International Electronics Symposium (IES)*. IEEE, 371–378.

[16] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. Focal loss for dense object detection. In *Proceedings of the IEEE international conference on computer vision*. 2980–2988.

[17] Yadini Pérez López, CFF Costa Filho, LMR Aguilera, and MGF Costa. 2017. Automatic classification of light field smear microscopy patches using Convolutional Neural Networks for identifying Mycobacterium Tuberculosis. In *2017 CHILEAN conference on electrical, electronics engineering, information and communication technologies (CHILECON)*. IEEE, 1–5.

[18] Aeri Rachmad, Nur Chamidah, and Riries Rulaningtyas. 2020. Mycobacterium tuberculosis images classification based on combining of convolutional neural network and support vector machine. *Commun. Math. Biol. Neurosci.* 2020 (2020), Article–ID.

[19] Amir Rasouli and John K Tsotsos. 2017. The effect of color space selection on detectability and discriminability of colored objects. *arXiv preprint arXiv:1702.05421* (2017).

[20] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. 2015. U-net: Convolutional networks for biomedical image segmentation. In *International Conference on Medical image computing and computer-assisted intervention*. Springer, 234–241.

[21] P Sathish, D Preethi, Clara Shanthi Dominic, and G Kadiravan. 2024. Hybrid Optimized Deep Learning-Based Bacilli Segmentation and Infection-Level Identification of Tuberculosis Using Sputum Images. *International Journal of Pattern Recognition and Artificial Intelligence* 38, 01 (2024), 2357017.

[22] Ramprasaath R Selvaraju, Michael Cogswell, Abhishek Das, Ramakrishna Vedantam, Devi Parikh, and Dhruv Batra. 2017. Grad-cam: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE international conference on Computer Vision*. 618–626.

[23] Inga G Shelomentseva and Serge V Chentsov. 2020. Classification of Microscopy Image Stained by Ziehl–Neelsen Method Using Different Architectures of Convolutional Neural Network. In *International Conference on Neuroinformatics*. Springer, 269–275.

[24] Christoph Spahn, Estibaliz Gómez-de Mariscal, Romain F Laine, Pedro M Pereira, Lucas von Chamier, Mia Conduit, Mariana G Pinho, Guillaume Jacquemet, Séamus Holden, Mike Heilemann, et al. 2022. DeepBacs for multi-task bacterial image analysis using open-source deep learning approaches. *Communications Biology* 5, 1 (2022), 688.

[25] Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. 2016. Rethinking the inception architecture for computer vision. In *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2818–2826.

[26] Catherine Vilchèze and Laurent Kremer. 2017. Acid-fast positive and acid-fast negative Mycobacterium tuberculosis: The Koch paradox. *Microbiology spectrum* 5, 2 (2017), 10–1128.

[27] Sanghyun Woo, Jongchan Park, Joon-Young Lee, and In So Kweon. 2018. CBAM: Convolutional Block Attention Module. In *European Conference on Computer Vision (ECCV)*. 3–19.

[28] World Health Organization. 2024. *Global tuberculosis report 2024*. Technical Report. World Health Organization. https://www.who.int/teams/global-tuberculosis-programme/tb-reports/global-tuberculosis-report-2024 [Online; accessed 30-Dec-2024].

[29] Sabina Zurac, Cristian Mogodici, Teodor Poncu, Mihai Trăscău, Cristiana Popp, Luciana Nichita, Mirela Cioplea, Bogdan Ceachi, Liana Sticlaru, Alexandra Cioroianu, et al. 2022. A new artificial intelligence-based method for identifying mycobacterium tuberculosis in Ziehl–Neelsen stain on tissue. *Diagnostics* 12, 6 (2022), 1484.